Bakal Goodhue-Cocom

goodhue-cocom.1@osu.edu | bgood20@illinois.edu | linkedin.com/in/bakal-goodhue-cocom/

Education

The Ohio State University **August 2026 - Expected May 2028**
B.S in Mechanical Engineering
Relevant Coursework: Dynamics, Linear Algebra, Statistical Analysis, Fundamentals of Engineering, Mechanics of Materials

University of Illinois Urbana-Champaign **August 2023 – May 2026**
B.S in Chemistry **GPA: 3.3**
Relevant Coursework: Solid Mechanics, Statics, Intro to CAD, Electrical and Electronic Circuits, Differential Equations

Skills

- 3+ years experience with Creo, ANSYS, SOLIDWORKS, Autodesk Fusion 360, Python, MATLAB, GD&T, CFD
- 3+ years hands-on experience with Formula SAE, Composites Manufacturing, 3D Printing, TIG Welding, CNC

Technical Experience

Formula Buckeyes – Formula SAE **Columbus, OH**
Powertrain Engineer **August 2026 – Present**

- Led a 4–6 week SolidWorks redesign of a two-year-stalled FSAE EV 600V, 9 kWh battery pack from the ground up
- Designed and packaged accumulator segments, contactor assemblies, HV/LV enclosures, busbar interfaces, and service features around fixed chassis and drivetrain constraints.
- Drove DFM, HV safety, harness routing, serviceability, and inverter integration decisions across a 5-person ME/EE team under a one-month vehicle deadline, handling design, manufacturing, debugging, integration and installation

United Airlines **San Francisco, CA**
Powerplant Engineering Intern - SFOEP **January 2026 – July 2026**

- Developed 30+ specialized repairs for non-standard damage on PW4000 parts, avoiding $11,047,493 in scrap
- Saved 22 PW4090 1st Stage LPC Blades from scrap after identifying a unlisted OEM approved repair within limits
- Revised 500+ job instruction cards to improve procedural accuracy, engineering controls, safety, and compliance
- Performed root cause analyses on engine and component failures, documenting findings and creating dispositions
- Provided technical support for IAE V2500 and PW4000 engine families across line and base maintenance operations

Cummins Inc. **Stoughton, WI**
Product Engineering Intern – Decomposition Reactor Development **June 2025 – December 2025**

- Led design, development and implementation of a universal diesel exhaust emissions test rig for an OEM's Euro VII, Euro VI aftertreatment systems, utilizing Creo and ANSYS for structural, thermal & CFD analysis
- Reduced rig manufacturing time by 6 weeks, saving $50,000 over original rig concepts and enhancing variable control
- Designed a modular hydrocarbon sampling rig to measure thermal & fuel efficiency of diesel oxidation catalysts
- Created a DEF deposit cleanout time function based on real-world test data to model downtime for X15 AT systems
- Developed AT test data analysis tool functions in MATLAB, including automated data reports and processing

Illini Electric Motorsports – Formula SAE **Champaign, IL**
High Voltage Integration and Chassis Project Lead **September 2023 – May 2025**

- Led manufacturing of carbon fiber accumulator/battery structures, cutting 2 months from production schedule
- Designed and manufactured jigs, for inverter manufacturing, accumulator molds, accumulator case welding, and more
- Designed critical suspension brackets in Creo, conducted FEA analysis using ANSYS for all loading scenarios
- Reduced bracket installation time by 55%, improved structural performance & reliability, cut machining costs by 65%
- Designed, manufactured, and tested an IP-68 Low Voltage enclosure, improving packaging and installation time

ChatCAD – Startup/Project **Champaign, IL**
Head of Research and Development **September 2024 – January 2025**

- Co-founded an AI CAD copilot integrated into Autodesk Fusion 360, won second place in hackathon
- Conducted 100+ tests and evaluations to identify strengths/weaknesses and to improve AI algorithms
- Collaborated with engineers at Foxconn to find use cases for ChatCAD implementation in their component design cycle
- Piloted ChatCAD at Illinois Makerlab, to determine feasibility for CAD novices to design 3D printed parts/products